

Foundry-Constrained Yield-Aware Design of Silicon Grating Couplers

One device, one process, derived math, runnable code. C-band TE grating coupler on 220 nm SOI.

13 September 2026

No devices have been fabricated. No 3-D FDTD inverse-design loop has been run. Those are the next experiments, not claims of this document. Every number below is produced by analysis/run_study.py and stored in analysis/results/study.json.

Abstract

We reformulate C-band TE fiber-to-chip grating-coupler design as a foundry-constrained, chance-constrained electromagnetic problem on a single public process: 220 nm SOI, 70 nm shallow etch, 2 μm BOX, 150 nm minimum CD, 8° SMF-28 in oxide. The Maxwell problem is non-convex. PDK rules are hard constraints. Yield is the probability that insertion loss at 1550 nm stays below a specification under published within-wafer variation. The experimental unit is a device, not a wafer.

A 1-D slab plus leaky-wave model, calibrated so a uniform grating matches Bozzola et al. (2015) at 53.7% 2-D FDTD, gives period $\Lambda = 622$ nm at 50% fill ($n_{\text{eff}} = 2.694$); DRC fill window [0.241, 0.759]; linear fill apodization to **2.26 dB** nominal (uniform 2.70 dB); process Monte Carlo mean **2.42 dB**, $\sigma = 0.15$ dB; **yield at IL ≤ 0.5 dB is zero**; yield at 2.5 dB is 78.7%. Thermo-optic walk $d\lambda/dT = 0.067$ nm/K; extra IL after a 20 K swing at fixed laser wavelength is 0.003 dB, not 0.01 dB/°C. On a 15 pJ/bit module with 25% laser share, beating a 2.8 dB PDK cell by 0.54 dB at both ends saves 5.5% of module energy.

A 0.3 dB / 70% yield target sits below the published -1.9 dB physics floor for 220 nm SOI without a back-reflector. That 65% figure is Bozzola’s published 2-D FDTD result, not a $\min()$ applied to this model. Tidy3D 2-D at exact 1550.00 nm, waveguide through the $-x$ PML, fiber aligned, 16 $\rightarrow$ 8 $\rightarrow$ 4 nm: uniform still wins (2.66 dB at 16 nm, 2.69 dB at 4 nm). Proposed drifts 2.76 $\rightarrow$ 2.99 dB under refinement. Fill-only is the worst 58 nm cell (3.69 dB). Sub-0.5 dB literature results use a metal mirror, a poly-Si overlay, unidirectional guided resonances, or thicker SOI. The honest production target on this stack is to beat a ~ 2.8 dB PDK grating toward the ~ 1.9 – 2.3 dB DRC-limited floor, with a 2.5 dB yield spec, on one MPW of paired loopbacks.

1. Scope

This report picks one device and one public process: a C-band TE grating coupler on 220 nm SOI. It writes the optimization problem with fields, chance constraints, and DRC; computes what 2-D FDTD can compute; and sizes an MPW experiment that exists on a price list. Non-goals: quantum, cryogenic, datacenter TCO, and any claim that this pipeline has been validated in silicon.

2. Device and process

220 nm SOI with a ~ 70 nm shallow etch is the default of AIM Photonics, imec iSiPP, AMF, and Cornerstone. It is a photonics MPW stack, not a logic CMOS node.

Quantity	Value	Note
Si device / BOX / etch	220 nm / 2.0 μm / 70 nm	photonics-grade SOI

Quantity	Value	Note
Cladding	SiO ₂ , n = 1.444	1550 nm
Fiber	SMF-28, MFD 10.4 μm, 8°	angle in the cladding
Polarization / λ	TE / 1550 nm	PDK TE vertical coupler class
Min CD / space	150 nm	193 nm DUV, conservative
AIM PDK TE GC	~2.8 dB, ~30 nm 1 dB BW	Fahrenkopf / Analog Photonics
imec iSiPP200 GC	< 2 dB (platform note)	different PDK, same idea

2.1 What “< 0.5 dB” actually requires

Bozzola et al., Opt. Express 23, 16289 (2015), put the 2-D FDTD ceiling for an apodized grating on **220 nm SOI with no back-reflector at 65% (−1.9 dB)**. The same paper reaches −0.5 dB only by moving to 340 nm SOI. Measured sub-decibel devices use a backside metal (Benedikovic 2015, −0.69 dB), topological unidirectional resonances on 340 nm SOI (Wang, Sci. Adv. 2024, −0.34 dB), a three-wave-interaction foundry coupler (Valdez 2025, −0.69 dB), or a simulated inverse design with a bottom reflector (Huang & Barz 2025, −0.35 dB). A production spec of 0.3 dB on standard 220 nm SOI, no mirror, is not ambitious. It is forbidden by the stack.

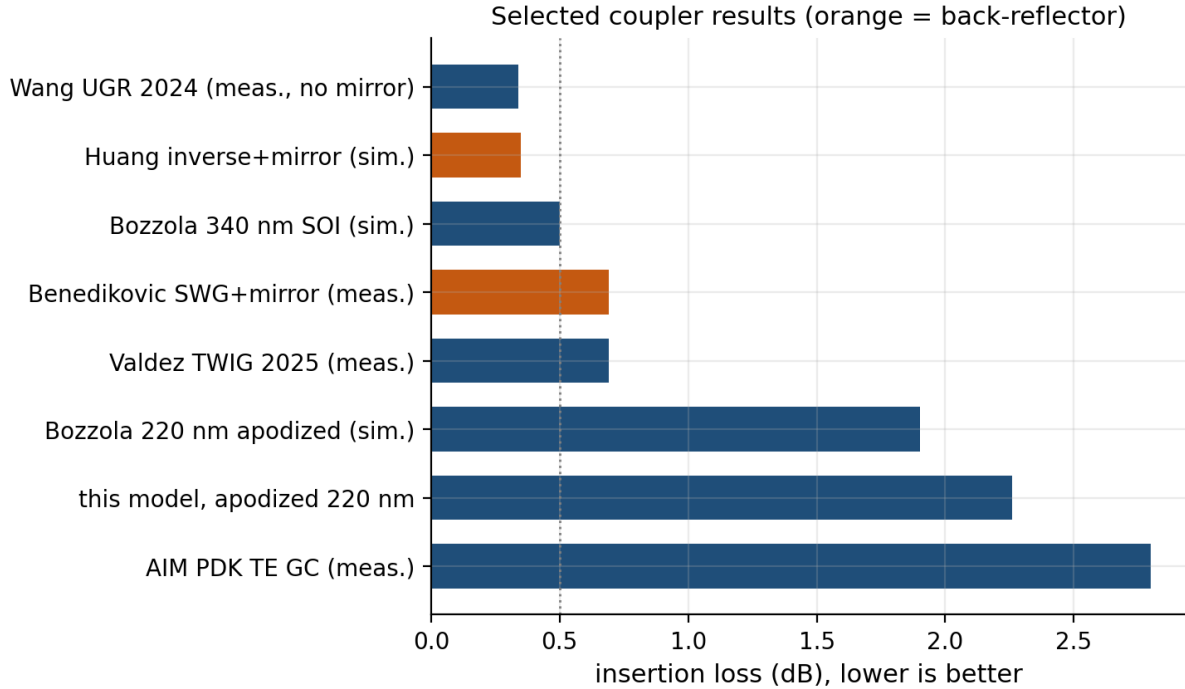


Figure 1. Selected coupler results. Orange bars use a back-reflector.

3. Electromagnetic model

Unetched (220 nm) and shallow-etched (150 nm) sections are symmetric oxide/Si/oxide slabs. Even TE modes satisfy $\kappa \sin(\kappa d/2) - \gamma \cos(\kappa d/2) = 0$, with $\kappa = k_0 \sqrt{(n_{\text{Si}}^2 - n_{\text{eff}}^2)}$ and $\gamma = k_0 \sqrt{(n_{\text{eff}}^2 - n_{\text{ox}}^2)}$. $n_{\text{Si}} = 3.476$, $n_{\text{ox}} = 1.444$ at 1550 nm. Unetched: $n_{\text{eff}} = 2.848$, $n_g = 3.577$, $\Gamma_{\text{Si}} = 0.810$. Etched: $n_{\text{eff}} = 2.539$, $n_g = 3.401$, $\Gamma_{\text{Si}} = 0.655$.

The Bloch index is the duty-cycle average $n_B(f) = f n_{\text{unetch}} + (1-f) n_{\text{etch}} = 2.694$ at $f = 0.5$. First-order phase match $\Lambda = \lambda / (n_B - n_c \sin \theta)$ gives $\Lambda = 622 \text{ nm}$ at 8° . Tooth and groove both $\geq 150 \text{ nm}$ force $f \in [0.241, 0.759]$. Apodization that asks for $f = 0.05$ is not a PDK-legal design.

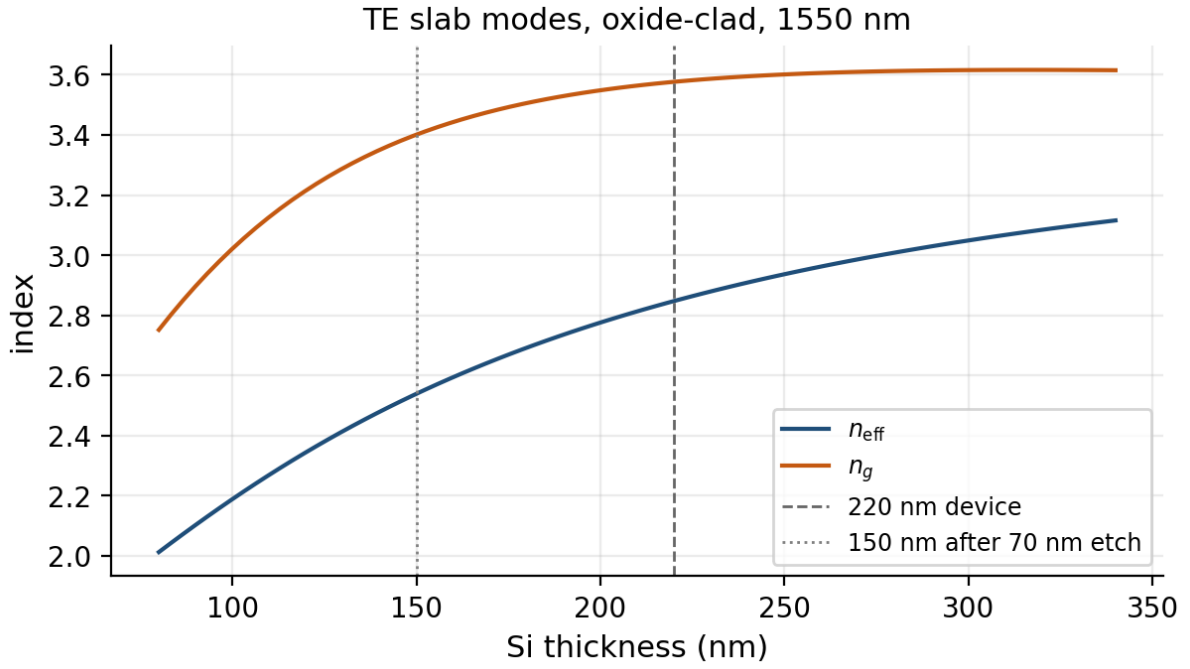
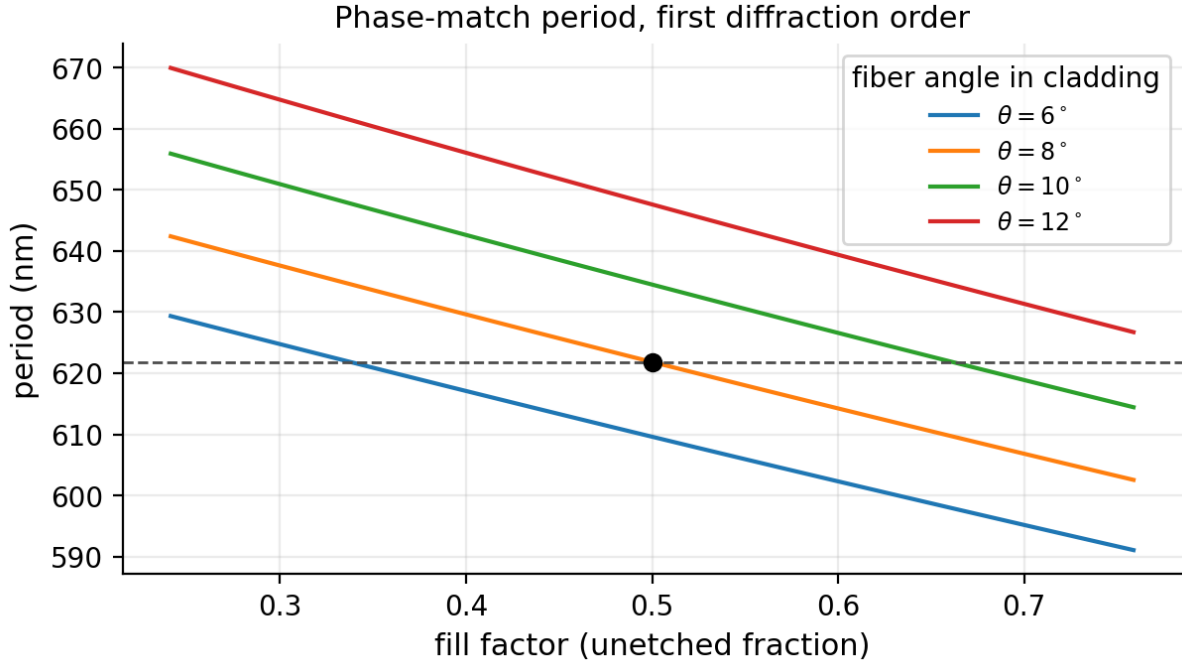

 Figure 2. TE slab n_{eff} and n_g versus Si thickness.


Figure 3. Phase-match period versus fill and fiber angle.

3.1 Coupling budget

Following Taillaert / Roelkens / Benedikovic, $\eta = D \cdot \eta_{\blacksquare} \cdot \eta_{2D} \cdot (1-R) \cdot \eta_{\text{taper}}$. D is upward directionality from a two-beam BOX/handle reflection model: $D = 0.725$ at $2.0 \mu\text{m}$ BOX. $\eta_{\blacksquare}$ is the overlap of the radiated near-field with the SMF-28 Gaussian ($w_0 = 5.2 \mu\text{m}$). Uniform overlap 0.870; linear fill apodization $0.759 \rightarrow 0.241$ over $12 \mu\text{m}$ raises it to 0.963. $\eta_{2D} = 0.871$ is a single calibration constant that puts the *uniform* device on Bozzola's 53.7% 2-D FDTD point. Second-order Bragg reflection is 0.33% at 8° (the reason the fiber is tilted). $\eta_{\text{taper}} = 0.98$.

Design	η	IL
Uniform, 50% fill (calibrated)	53.7%	2.70 dB
Linear apodization, DRC-legal	59.4%	2.26 dB
Bozzola unconstrained 220 nm FDTD ceiling	65%	1.90 dB
AIM PDK TE GC (measured)	~52%	~2.8 dB

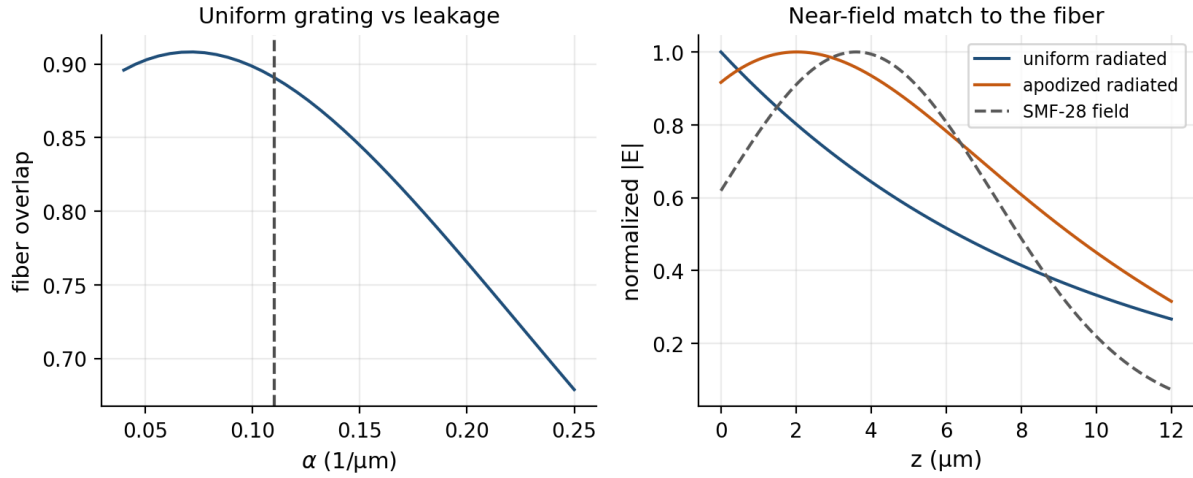


Figure 4. Uniform leakage overlap and radiated near-field versus SMF-28.

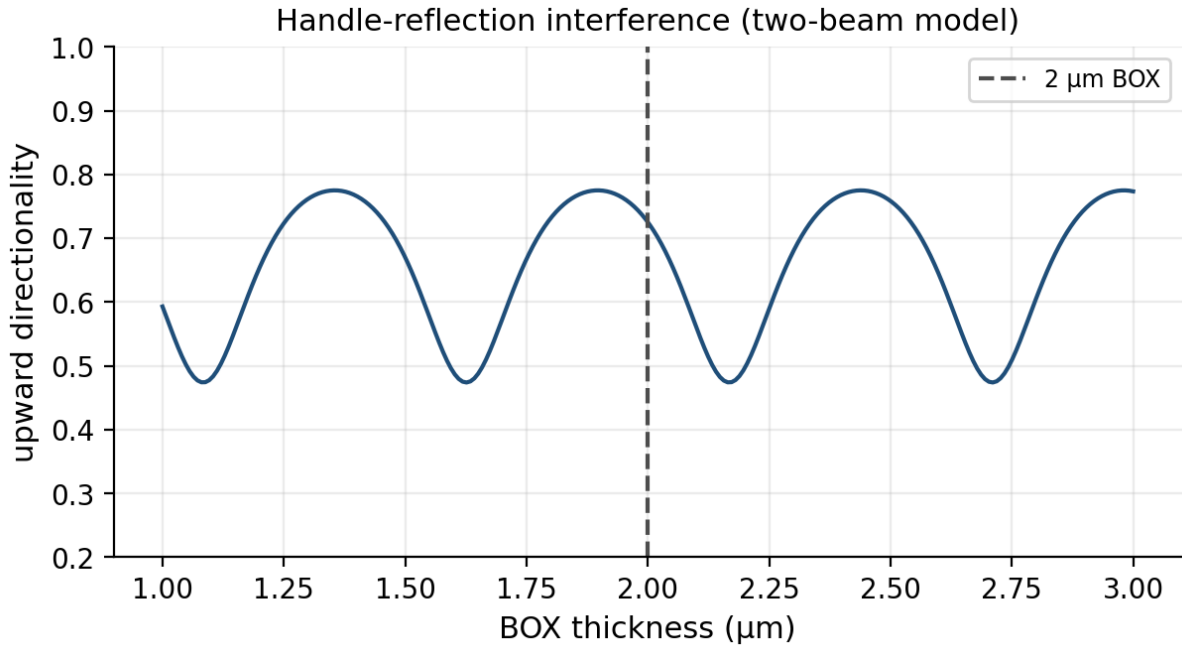


Figure 5. Upward directionality versus BOX thickness (two-beam model).

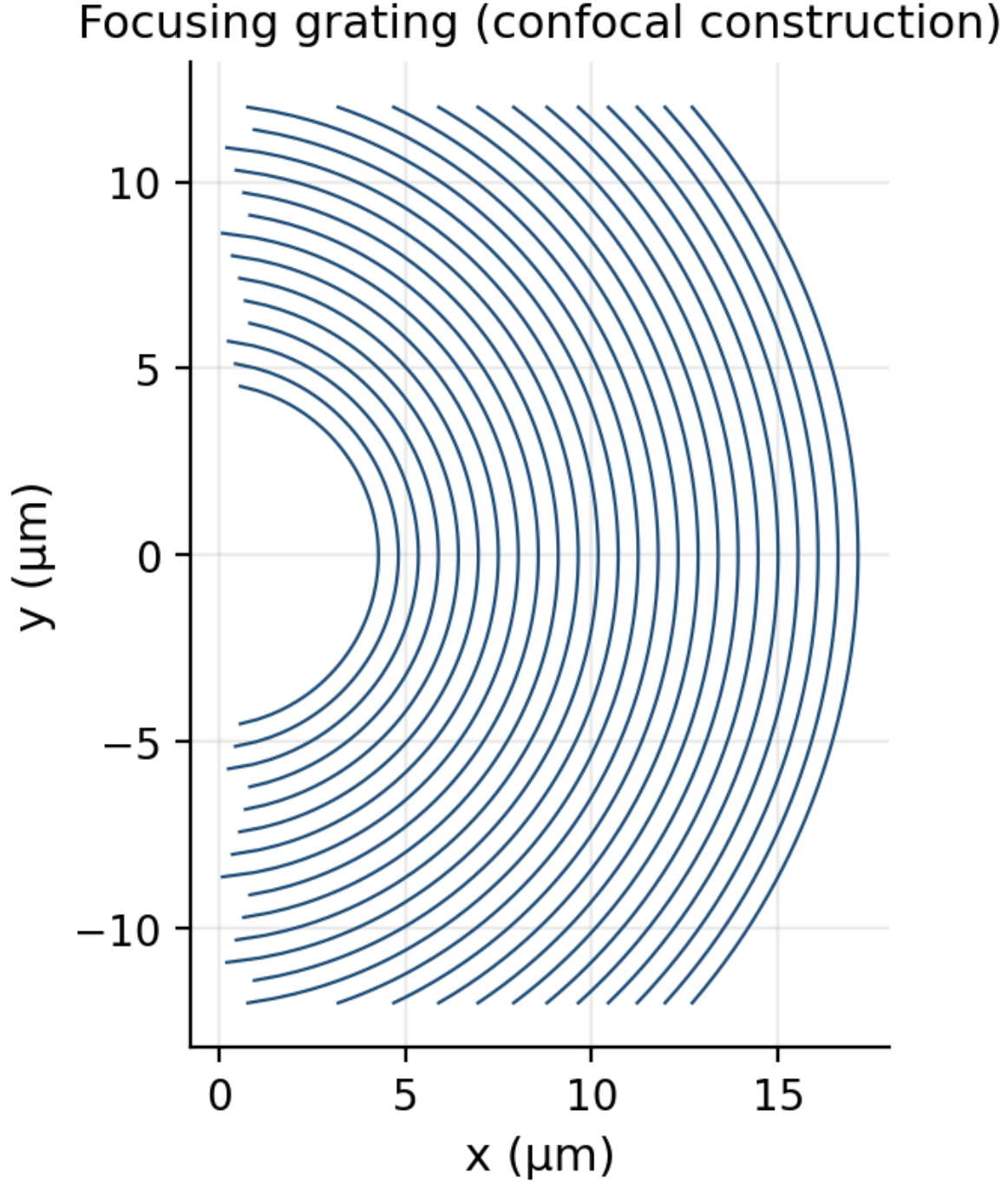


Figure 6. Focusing grating from the confocal construction. This is the GDS, not a 1-D bar array.

This model is not a substitute for 2-D/3-D FDTD. It does not capture the exact radiation angle versus etch or substrate higher-order reflections. It is enough to size the period and DRC window, show that 0.5 dB is impossible on this stack, propagate process error into yield, and write a correct optimization problem for the FDTD loop.

4. The optimization problem, written correctly

Let x be the design (fill samples, spline controls, or a filtered density), p the process vector, and $\eta(x, p; \lambda)$ the fiber overlap of the Maxwell field. $IL = -10 \log_{10} \eta$. The problem we actually want is: minimize $E_p[IL(x, p; \lambda_0)]$ subject to $P_p(IL \leq IL_{\text{spec}}) \geq Y_{\text{spec}}$, $BW_{1dB} \geq 25 \text{ nm}$, $R_{\text{back}} \leq 10^{-1.5}$, and $DRC(x) = \text{true}$. Recommended on this stack: $IL_{\text{spec}} = 2.5$

dB, $Y_{\text{spec}} = 0.80$. Not 0.3 dB and 70%.

A weighted sum $f = \omega_1 L - \omega_2 Y + \omega_3 T + \omega_4 P$ is not the problem we want. L , Y , T , P have different units. FDTD coupler loss is non-convex, so KKT is not a solution method. Yield is a discontinuous functional of process. PDK rules are indicators, not a soft fourth term.

A tractable surrogate (in `optimize.py`) is sample-average approximation plus CVaR on the upper tail of IL, with $x = (f_{\text{start}}, f_{\text{end}}, L)$ inside the DRC box. A Nelder–Mead pass moved mean IL from 2.42 dB to 2.35 dB and 2.5 dB yield from 79% to 86%. That is a real, small gain. The FDTD loop should solve the same problem with x a filtered density. Adjoint gradients (Lalau-Keraly et al., Opt. Express 21, 21693, 2013) give $\partial J / \partial \epsilon(r)$ from one forward and one adjoint Maxwell solve. Min CD enters by a 75 nm conic filter plus Heaviside projection (Piggott, fabrication-constrained inverse design, 2017). We have not run that loop. The design spec says how.

5. Thermal behaviour, derived rather than asserted

Differentiating the phase-match condition at fixed angle, with $n_g = n_{\text{eff}} - \lambda \partial n_{\text{eff}} / \partial \lambda$, gives $d\lambda/dT = \lambda (\partial n_{\text{eff}} / \partial T + n_{\text{eff}} \alpha_{\text{Si}}) / (n_g - n_c \sin \theta)$. Using Komma et al., Appl. Phys. Lett. 101, 041905 (2012), $dn_{\text{Si}}/dT = 1.80 \times 10^{-4} / \text{K}$: $d\lambda/dT = 0.067 \text{ nm/K}$. A grating coupler is broadband. Extra IL at the design wavelength after a peak walk $\delta\lambda$ on a Gaussian spectrum is $12.04 (\delta\lambda/\text{FWHM})^2 \text{ dB}$. A 20 K swing walks the peak by 1.33 nm and costs **0.003 dB**. The linear coefficient “0.01 dB/°C” is the wrong functional form (detuning loss is quadratic in ΔT) and is two orders of magnitude too large on this device class.

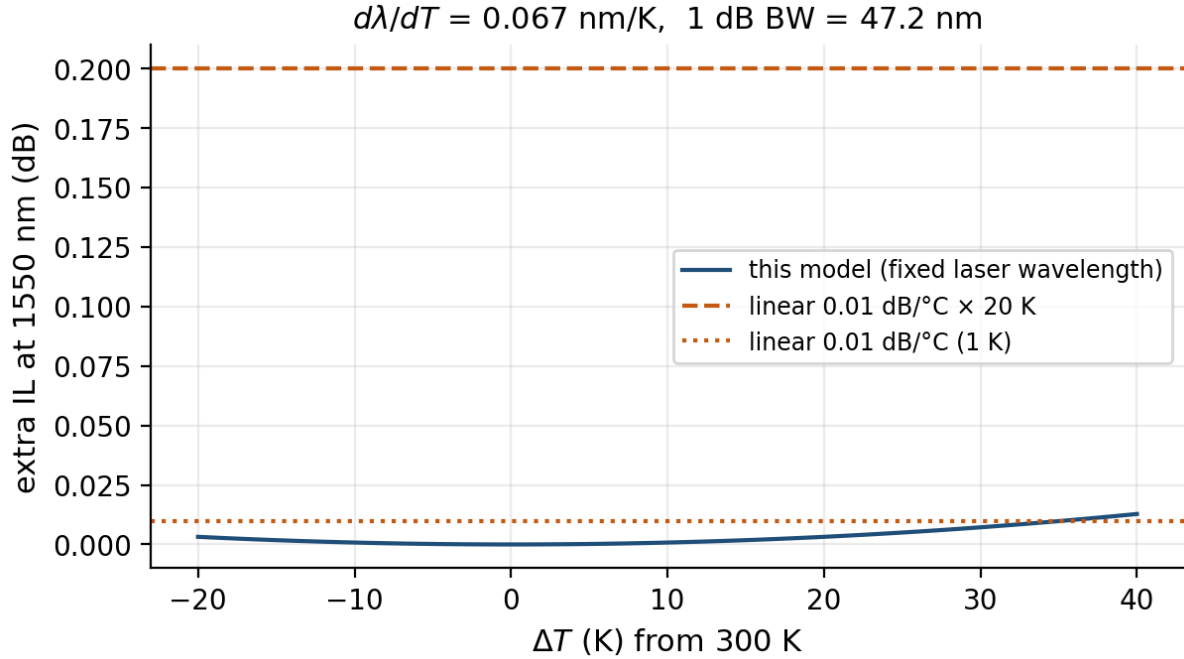


Figure 7. Extra IL at 1550 nm versus temperature. A linear 0.01 dB/°C $\times$ 20°C line would be 0.20 dB; the model is quadratic and much smaller.

6. Yield

Within-wafer 1-sigma values from 193 nm DUV silicon photonics (ACS Photonics 10, 928, 2023) and Chrostowski’s grating-coupler wavelength sensitivities (Silicon Photonics Design, Ch. 11): $\sigma(t_{\text{Si}}) = 2.0 \text{ nm}$ (1.82 nm/nm), $\sigma(\text{etch}) = 3.0 \text{ nm}$ (1.90 nm/nm), $\sigma(\text{CD}) = 2.6 \text{ nm}$ (0.215 nm/nm), $\sigma(\theta) = 0.4^\circ$, $\sigma(\text{offset}) = 0.8 \mu\text{m}$. IL at 1550 nm is peak IL of the perturbed geometry plus detuning plus a small angular penalty. This is within-wafer scatter, not wafer-to-wafer mean shift.

Statistic	Value
Nominal apodized IL	2.26 dB
Monte Carlo mean / median / σ	2.42 / 2.38 / 0.15 dB
5th / 95th percentile	2.27 / 2.74 dB
$P(IL \leq 0.5 \text{ dB})$	0
$P(IL \leq 2.0 \text{ dB})$	0
$P(IL \leq 2.5 \text{ dB})$	78.7% (95% CI 77.0–80.3%)
$P(IL \leq 3.0 \text{ dB})$	99.4%

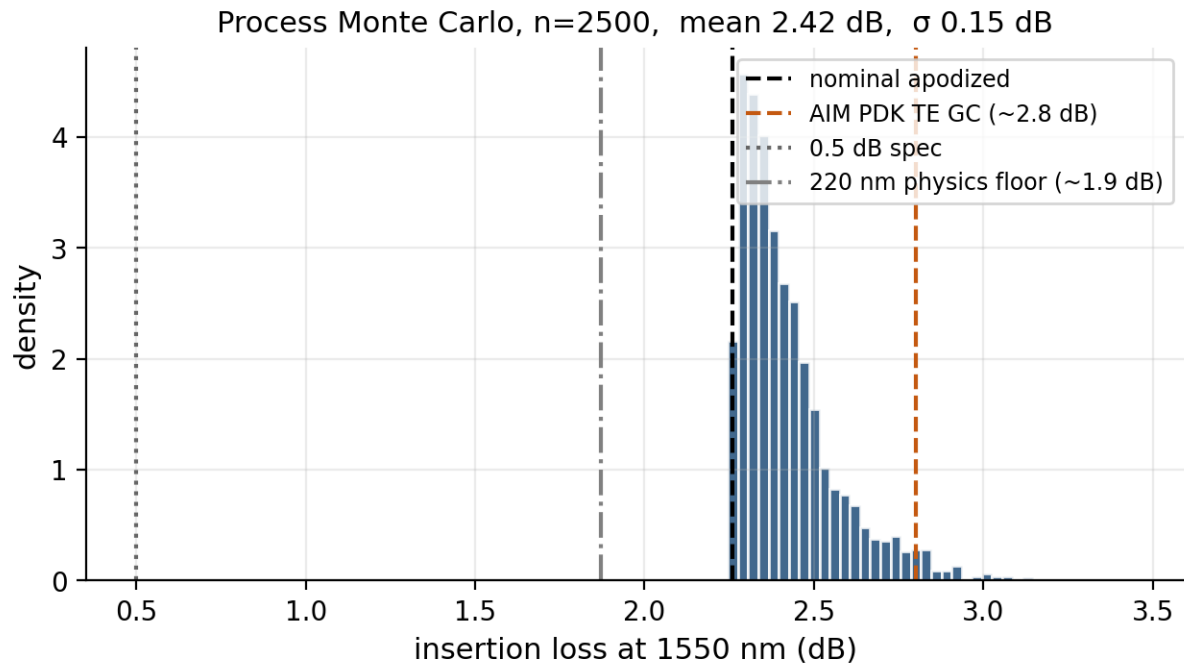


Figure 8. Process Monte Carlo of IL at 1550 nm. The 0.5 dB target is empty.

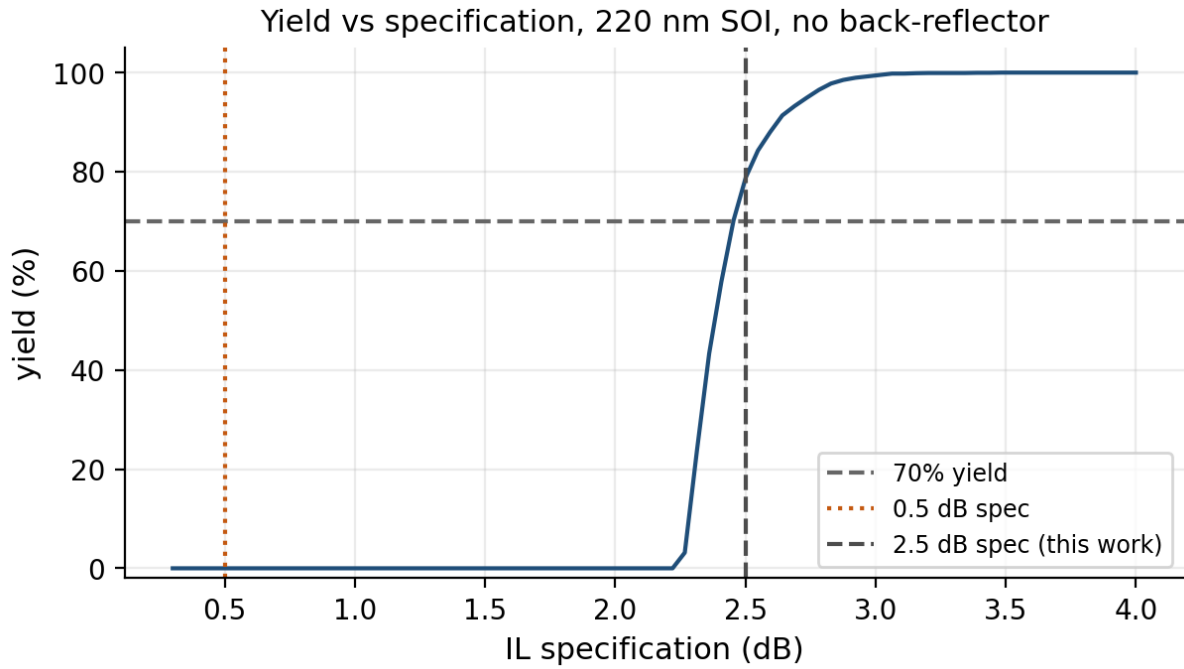


Figure 9. Yield versus IL specification.

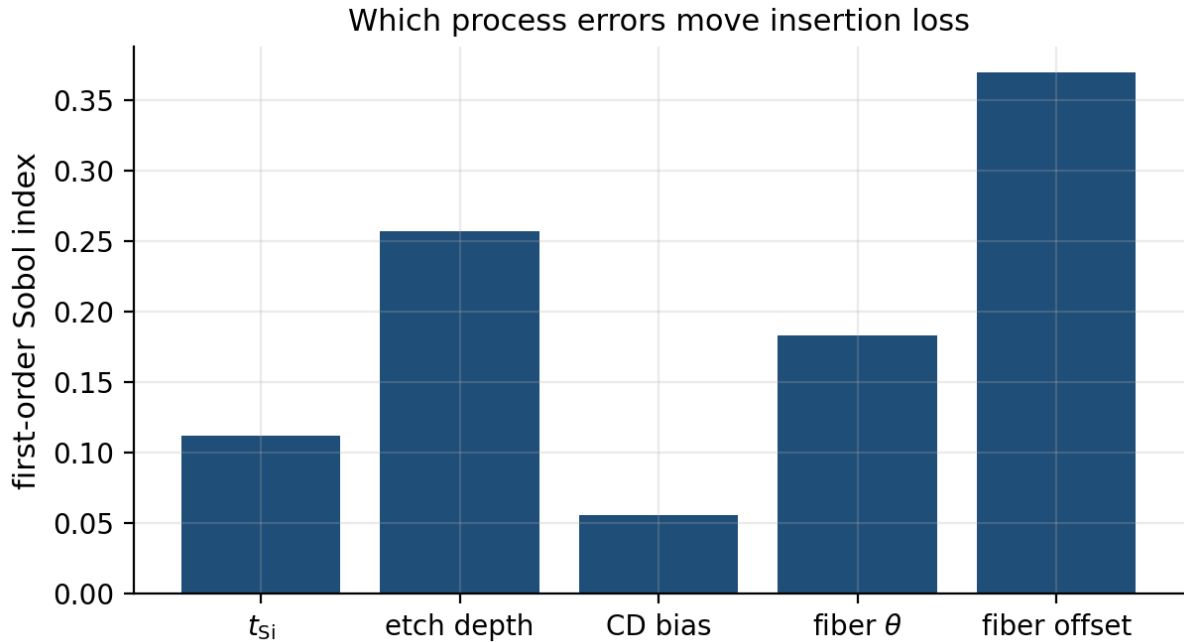


Figure 10. Jansen first-order Sobol indices of IL. Alignment and etch dominate.

The 0.5 dB target has yield zero because the *uncapped* leaky-wave mean is 2.42 dB and Bozzola’s published 2-D FDTD ceiling on this stack is 1.9 dB. Neither is a Python `min()` that copies 65% into the Monte Carlo. A 2.5 dB specification — 0.3 dB better than the AIM PDK cell, 0.25 dB worse than our mean — is the spec that has a chance of being real. Jansen first-order Sobol indices: fiber offset 0.37, etch 0.26, angle 0.18, t_{Si} 0.11, CD 0.06 (sum 0.98).

7. Energy, without the category error

An 800G pluggable is ~15 pJ/bit, of which the laser electrical share is ~20–30%. Copper DAC energy is a reach comparison. Coupler IL does not convert copper into photonics. Laser optical power scales as $10^{(n_c \cdot \Delta \text{IL}/10)}$ for n_c

couplers in the laser-to-PD path. For a packaged TX+RX pair, $n_c = 2$.

Change	Laser scale	% of 15 pJ/bit module	Saved
PDK 2.8 → this 2.26 dB	1.28×	5.5%	0.83 pJ/bit
PDK 2.8 → 0.5 dB (if it existed)	2.88×	16.3%	2.45 pJ/bit
3.0 → 0.3 dB (if it existed)	3.47×	17.8%	2.67 pJ/bit

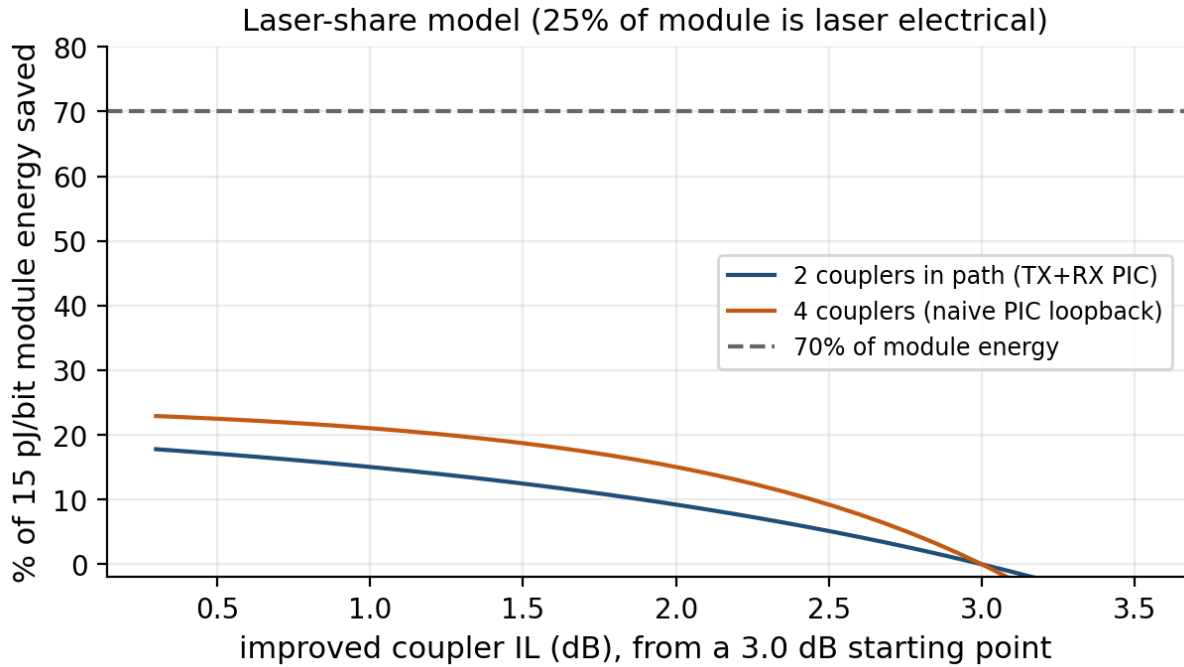


Figure 11. Module-energy fraction saved versus improved coupler IL. A 70% module cut is not available from the coupler.

A 70–90% reduction versus copper does not follow from a coupler. Sub-pJ/bit transmitters exist; they get there by killing SERDES and DSP. A coupler improvement is worth about one pJ/bit on a pluggable.

8. An experiment that can be purchased

Yield of a coupler is measured on devices, not wafers. A 25 mm² AIM die holds hundreds of 40 μm $\times$ 40 μm focusing couplers. Textbook sizes, $\alpha = 0.05$, 80% power:

Test	Effect	n per arm
Two-proportion yield 50% vs 70%	20 points	91 devices
Two-sample t, $\Delta\text{IL} = 0.30$ dB, $\sigma = 0.30$ dB	mean IL	16 devices
Paired t, $\sigma_{\text{diff}} = 0.20$ dB	0.30 dB	4 pairs

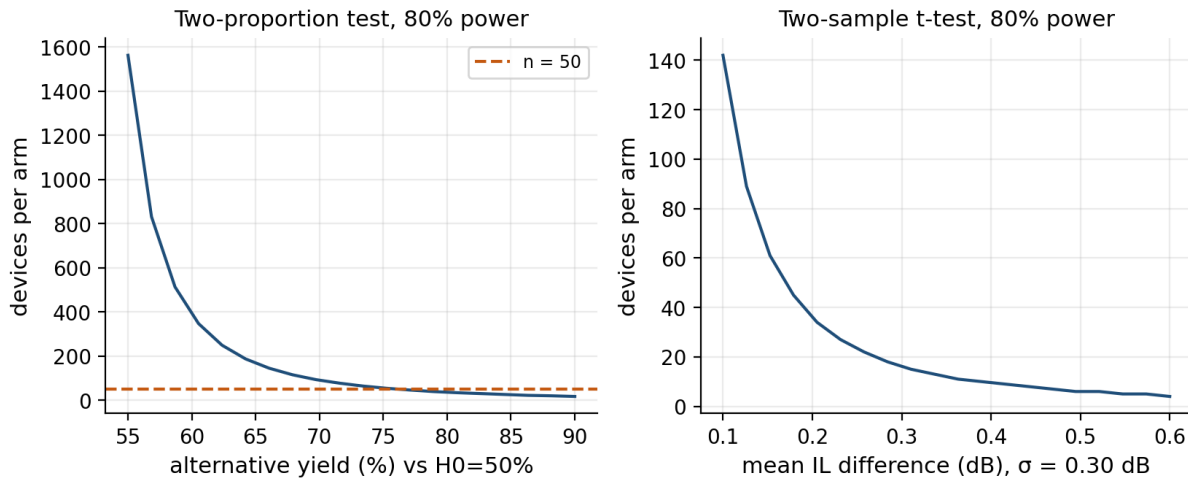


Figure 12. Device-level sample size for a yield or mean-IL test.

Put 100 PDK loopbacks and 100 candidate loopbacks on one reticle, plus a period/fill ladder and PCM. AIM Photonics member price for 20 chips of a 25 mm² passive PIC is \$28,600 at the published volume table (accessed 2026-09-13). Masks are in the MPW price. This is the \$30k experiment. Falsification: if the candidate’s paired mean IL is not ≥ 0.20 dB better than the PDK cell with a 95% interval excluding zero, the design claim is false on this process — and that is publishable.

9. What to do next, in order

1. Done: Tidy3D re-run with waveguide through the $-x$ PML, matched x_{fib} , fill-only on the etch sweep, 16 $\rightarrow$ 8 $\rightarrow$ 4 nm. Uniform 2.69 dB at 4 nm; proposed drifts to 2.99 dB. Fill-only is the worst 58 nm cell. Do not start adjoint on the 16 nm apodized ranking.
2. Adjoint inverse design with a 75 nm filter, 150 nm projection, 8° fiber, CVaR on a 20-point process batch. Compare to the linear apodization. That is a paper if you beat the PDK cell in FDTD and on the MPW.
3. One MPW, paired loopbacks, as in `design/mpw-protocol.md`. That is the experiment that can be bought.
4. Only then: poly-Si overlay or a foundry-legal backside mirror, if the PDK has them, if you want sub-1 dB. Do not claim inverse-design validation before the grid has converged and an MPW exists.

Selected references

- A. Bozzola et al., Opt. Express 23, 16289 (2015). 220 nm no-mirror ceiling -1.9 dB; 340 nm -0.5 dB.
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D. Taillaert, P. Bienstman, R. Baets, Opt. Lett. 29, 2749 (2004). Compact SOI grating coupler.
Full citation list: `paper/manuscript.md`.

Canonical source: `paper/manuscript.md`. Design spec: `design/design-spec.md`. MPW protocol: `design/mpw-protocol.md`. Reproduce: `python analysis/run_study.py`.